

Lecture 33

Rejection Sampling

Ref Sheldon M. Ross
Chapter 10.

You want to get samples of a continuous R.V with pdf $f(x)$, $x \in \mathbb{R}$ and there exists another pdf

$g(x)$, $x \in \mathbb{R}$ st $f(x)$ *hard*

easy $f(x) \leq c g(x)$, $x \in \mathbb{R}$.

where c is some constant

① $U \sim \text{Uniform}[0, 1]$,

$Y \sim g(\cdot)$. [assume U, Y are independent]

② If $U \leq \frac{f(Y)}{c g(Y)}$ accept the

Recap.

Inverse sampling.

If CDF $F(\cdot)$ is known for a random variable X and if we want to get samples of X then

$$X = F^{-1}(U).$$

is a way of getting samples. given that F is an invertible function.

Gaussian samples:

$X \sim N(0, 1)$. X, Y are
 $Y \sim N(0, 1)$ independent

$\Downarrow$

$$R, \theta, \quad R = \sqrt{X^2 + Y^2} \\ \theta = \tan^{-1}(Y/X)$$

$c g(Y)$ sample Y
i.e., set $x = Y$.

Otherwise repeat from step 1.

are independent of each other.

Let $(U_1, Y_1), (U_2, Y_2), \dots$ be the
samples collected in N steps where

$$N = \min \left\{ n : U_n \leq \frac{f(Y_n)}{c g(Y_n)} \right\}.$$

$$X = Y_N.$$

We want to show that $F_X(x) = \int_{-\infty}^x f(x) dx$

$$F_X(x) = P(X \leq x) = P(Y_N \leq x).$$

$$= \sum_n P(N=n) P(Y_n \leq x | N=n)$$

$$= \sum_n P(N=n) P(Y_n \leq x | \underbrace{U_i > \frac{f(Y_i)}{c g(Y_i)}; U_n \leq \frac{f(Y_n)}{c g(Y_n)}}_{i=1, 2, \dots, n-1})$$

$$= \sum_n P(N=n) P(Y_n \leq x | U_n \leq \frac{f(Y_n)}{c g(Y_n)})$$

↓
won't be a function
of n

We know that U_i 's are
identical &
 Y_i 's are identical

$$= P(Y \leq x | U \leq \frac{f(Y)}{c g(Y)})$$

$$\sum_n P(N=n) = 1$$

$Y \sim g(\cdot)$
 $U \sim \text{Uniform}[0, 1]$

$$= \frac{P(Y \leq x, U \leq \frac{f(r)}{c g(r)})}{P(U \leq \frac{f(r)}{c g(r)})}$$

$$P(U \leq \frac{f(r)}{c g(r)}) =$$

$$\int_{-\infty}^{\infty} \int_0^{\frac{f(y)}{c g(y)}} g(y) \cdot 1 \, du \, dy$$

$$g(y) \cdot 1 \, du \, dy$$

Y and U are independent.

$$= \int_{-\infty}^{\infty} \cancel{g(y)} \frac{f(y)}{c \cancel{g(y)}} \, dy$$

$$= \frac{1}{c}$$

↓
This is the acceptance probability

always guaranteed to be ≤ 1 .

$$P\left(Y \leq x, U \leq \frac{f(y)}{c g(y)}\right) = \int_{-\infty}^x \int_0^{\frac{f(y)}{c g(y)}} g(y) \cdot 1 \, du \, dy$$

$$= \int_{-\infty}^x \cancel{g(y)} \frac{f(y)}{c \cancel{g(y)}} \, dy$$

$$= \frac{\int_{-\infty}^x f(y) \, dy}{c}$$

$$\Rightarrow F_X(x) = \frac{\int_{-\infty}^x f(y) \, dy}{\int_{-\infty}^{\infty} f(y) \, dy} = \int_{-\infty}^x f(y) \, dy.$$

$$\int_0^1 c_0 (x - x^2) \, dx = 1$$

$$c_0 \cdot \left[\frac{1}{2} - \frac{1}{3} \right] = 1$$

Example: $f(x) = c_0 x(1-x)$. $x \in [0, 1]$.

$$c_0 = 6$$

To find $g(x)$, c st

$$f(x) \leq c g(x) \quad x \in \mathbb{R} \rightarrow \textcircled{1}$$

$$g(x) = 1 \quad x \in [0, 1].$$

To find a constant c st eqn $\textcircled{1}$ is satisfied.

$$c_0 x(1-x) \leq c \cdot 1 \quad x \in [0, 1]$$

We know that

$$x(1-x) \leq \left[\frac{x + (1-x)}{2} \right]^2$$

AM-GM inequality.

Practice set asks you to find the marginal distribution of Y_i .

$X \sim \text{Beta}(\alpha, \beta)$

$$f_X(x) = \frac{x^{\alpha-1} (1-x)^{\beta-1}}{B(\alpha, \beta)} \quad x \in [0, 1]$$

Ordered statistics

$X_1, X_2, \dots, X_n$
are iid
Uniform $[0, 1]$

$Y_i = i\text{-th in ascending order of } x_1, \dots, x_n$

$$\Rightarrow C_0 \alpha (1-\alpha) \leq \frac{C_0}{4}.$$

$$\text{So pick } C = \frac{C_0}{4}.$$

To simulate this beta distribution we can directly use the uniform distribution.

$$\textcircled{1} \quad U \sim \text{Uniform}[0,1] \quad Y \sim \text{Uniform}[0,1].$$

$$\textcircled{2} \quad \text{If } U \leq \frac{f(Y)}{C} = \frac{Y(1-Y)}{4} \text{ then accept } Y. \\ \text{Otherwise repeat step 1.}$$

$$E[N] = C = \frac{C_0}{4} = \frac{6}{4} = 1.5.$$

$$N = \min \left\{ n : U_n \leq \frac{f(X_n)}{c g(X_n)} \right\}.$$

$$P\left(U_n \leq \frac{f(X_n)}{c g(X_n)}\right) = \gamma/c.$$

$$N \sim \text{Geometric}(\gamma/c).$$

$$E[N] = c.$$

$$X \sim \text{Geometric}(p)$$

$$E[X] = \gamma/p.$$

Convergence of R.V.s.

Ref: principles of Math Analysis
by Rudin

Recap of convergence of sequences and series

For a sequence of real numbers $a_1, a_2, \dots = (a_n)_{n \geq 1}$

$(a_n)_{n \geq 1}$ is said to converge to "a" if the following holds.

$$\forall \varepsilon > 0, \exists N(\varepsilon) \text{ st.}$$

$$|a_n - a| \leq \varepsilon.$$

$$n \geq N.$$

Example:

①

$$(a_n) = 1 \quad \frac{1}{2} \quad \frac{1}{3} \quad \frac{1}{4} \quad \frac{1}{5} \quad \dots$$

$$a_n = \frac{1}{n} \quad n \geq 1.$$

$$a_n \rightarrow 0.$$

$$\varepsilon = \frac{1}{6}.$$
$$n \geq N = 6.$$

$$\Rightarrow N\left(\frac{1}{6}\right) = 6.$$

for $\varepsilon = 10^{-6}$.

$$N(\varepsilon) = 10^6.$$

$$n \geq 10^6, |a_n - 0| \leq \varepsilon$$

② $a_n = (-1)^n$

doesn't converge.

$-1, 1, -1, 1, \dots$

Suppose a_n doesn't converge. $\Rightarrow$

$$\exists \epsilon > 0 \text{ st } \forall N \in \mathbb{N}, \exists n \geq N \\ |a_n - a| > \epsilon.$$

} negation
of
convergence
definition.

Formal proof:

Let $\epsilon = \frac{1}{100}$.

$\forall N \in \mathbb{N}$

if $|1 - a| > \frac{1}{100}$. then we

already have an example $n \geq N$
where the condition fails.

$\Leftrightarrow$

Suppose $|1 - a| \leq \frac{1}{100}$

$\Rightarrow |1 - a| = 1 + a > \frac{1}{100}$

then $|1 - a| > \frac{1}{100} \Rightarrow \exists n \geq N$ for
which $|a_n - a| > \epsilon$.

Properties of sequences

① $a_n \rightarrow a$, $b_n \rightarrow b$.

$$a_n + b_n \rightarrow a + b$$

$$a_n - b_n \rightarrow a - b.$$

② $l_n \leq a_n \leq u_n$.

$$\lim_{n \rightarrow \infty} l_n \leq \lim_{n \rightarrow \infty} a_n \leq \lim_{n \rightarrow \infty} u_n$$

③ Any monotonic sequence that is bounded converges.

$(a_n) = \frac{1}{n}$ ④ $a_1 \geq a_2 \geq \dots \geq l$. then.

$(a_n)_{n \geq 1}$ converges.

$(a_n) = 1 - \frac{1}{n}$ ⑤ $a_1 \leq a_2 \leq \dots \leq u$ then $(a_n)_{n \geq 1}$ converges.